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MLPY: A Modular and Extensible Machine Learning Framework for Python

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Abstract

Machine learning in Python is dominated by monolithic frameworks that, while powerful, lack the modularity and composability needed for modern ML workflows. We present MLPY, a modular and extensible framework inspired by mlr3's design principles, bringing object-oriented composability to Python's ML ecosystem. MLPY provides unified abstractions for tasks, learners, measures, and resampling strategies, enabling complex ML workflows with minimal code. Unlike existing frameworks, MLPY includes native support for spatial machine learning with spatially-aware cross-validation and automatic data validation. Benchmarks show MLPY achieves comparable performance to scikit-learn while providing a more modular and flexible API. The framework is open source under Apache 2.0 license and available at `https://github.com/franciscoparrao/mlpy`.

Keywords: Machine Learning, Python, Framework, Modularity, Spatial ML, MLOps, AutoML

1 Introduction

Machine learning has become a cornerstone of modern data analysis, with Python emerging as the dominant language for ML applications. However, current Python ML frameworks face significant limitations in modularity and composability.

1.1 Motivation

The scikit-learn library (Pedregosa et al., 2011), while excellent for basic tasks, adopts a monolithic design where preprocessing, modeling, and evaluation are tightly coupled. This makes it difficult to:

- Compose complex ML pipelines from reusable components
- Systematically compare multiple learners across multiple datasets
- Extend the framework with custom components without violating abstractions
- Handle specialized tasks like spatial ML or time series analysis

In contrast, the R ecosystem has successfully adopted modular designs. The mlr3 framework (Lang et al., 2019) demonstrates how object-oriented abstractions can simplify complex ML workflows while maintaining extensibility.

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1.2 Contributions

This paper presents MLPY, a modular ML framework for Python that makes the following contributions:

1. **Unified abstractions**: Task, Learner, Measure, and Prediction classes that separate concerns and enable composition
2. **Flexible pipelines**: Graph-based pipeline system supporting complex preprocessing workflows
3. **Spatial ML support**: First Python ML framework with native spatial cross-validation and spatial-aware measures
4. **Data validation**: Automatic validation system detecting common data quality issues
5. **Extensibility**: Clean extension points for custom tasks, learners, measures, and pipeline operations

The rest of this paper is organized as follows: Section 2 reviews related work; Section 3 describes MLPY's architecture; Section 4 discusses implementation details; Section 5 presents benchmarks; Section 6 shows usage examples; and Section 7 concludes.

2 Related Work

2.1 Python ML Frameworks

scikit-learn The scikit-learn library (Pedregosa et al., 2011) is the de facto standard for ML in Python. It provides a consistent API via the `fit`/`predict` interface and includes implementations of most common ML algorithms. However, its monolithic design makes it difficult to compose complex workflows. For example, comparing multiple models across multiple datasets requires significant boilerplate code (see Section 5).

PyCaret PyCaret (Ali, 2020) focuses on AutoML and low-code ML workflows. While it simplifies common tasks, it sacrifices flexibility and is less suitable for research or custom workflows.

H2O H2O (H2O.ai, 2021) is an enterprise ML platform with Python bindings. It excels at distributed computing but has a proprietary core and limited extensibility.

2.2 R ML Frameworks

mlr3 The mlr3 framework (Lang et al., 2019) pioneered the modular object-oriented approach that MLPY adopts. It provides clean abstractions for tasks, learners, measures, and resampling strategies. MLPY brings these design principles to Python while adding unique features like spatial ML support.

caret The caret package (Kuhn, 2008) was an early attempt at unified ML interfaces in R. However, it used a functional rather than object-oriented design, limiting composability.

tidymodels tidymodels (Kuhn and Wickham, 2020) is a modern R framework following tidyverse principles. It provides excellent integration with data manipulation tools but is R-specific.

2.3 Comparison

Table 1 compares MLPY with related frameworks.

Table 1: Comparison of ML frameworks

Framework	Lang	Modular	Spatial ML	Pipelines	AutoML	Open
scikit-learn	Python	No	No	Limited	No	Yes
PyCaret	Python	No	No	Yes	Yes	Yes
H2O	Multi	No	No	Yes	Yes	Mixed
mlr3	R	Yes	Yes	Yes	Yes	Yes
caret	R	Partial	No	Yes	Yes	Yes
tidymodels	R	Yes	No	Yes	Partial	Yes
MLPY	**Python**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**

MLPY is the only Python framework combining modularity, spatial ML, and extensibility.

3 Architecture and Design

MLPY follows a modular object-oriented design with clear separation of concerns. Figure **??** shows the overall architecture.

3.1 Core Abstractions

Task A Task encapsulates an ML problem with its data and metadata. It provides a consistent interface for accessing features, targets, and row indices regardless of the underlying data representation.

```
task = TaskClassif(data=df, target='species')
print(task.feature_names)  # ['sepal_length', 'sepal_width', ...]
print(task.target_names)   # ['species']
print(task.class_names)    # ['setosa', 'versicolor', 'virginica']
```

Task subclasses specialize for different problem types:

- `TaskClassif` / `TaskRegr` — supervised learning
- `TaskCluster` — unsupervised learning
- `TaskTimeSeries` / `TaskForecasting` — temporal data
- `TaskClassifSpatial` / `TaskRegrSpatial` — spatial data

Learner A Learner wraps an ML algorithm with a consistent interface for training and prediction. Unlike scikit-learn's stateful estimators, MLPY learners are designed to be cloned and reused safely.

```
learner = LearnerRandomForestClassifier(n_estimators=100)
learner.train(task)
predictions = learner.predict(task)
```

Prediction A Prediction object encapsulates model outputs with metadata. It stores ground truth, predicted values, and (optionally) prediction probabilities or standard errors.

Measure A Measure computes performance metrics from predictions. Measures are first-class objects that can be passed to resampling and benchmarking functions.

```
measure = MeasureClassifAccuracy()
score = measure.score(predictions.truth, predictions.response)
```

3.2 Resampling Strategies

MLPY provides a flexible resampling system for cross-validation and performance estimation:

- `ResamplingCV` — k-fold cross-validation
- `ResamplingHoldout` — train/test split
- `ResamplingBootstrap` — bootstrap resampling
- `SpatialKFold` — spatial cross-validation (prevents spatial leakage)
- `SpatialBlockCV` — block-based spatial CV
- `SpatialBufferCV` — buffer-based spatial CV

The `resample()` function provides a high-level interface:

```
result = resample(
    task=task,
    learner=learner,
    resampling=ResamplingCV(folds=5),
    measures=[MeasureClassifAccuracy(), MeasureClassifF1()]
)
print(result.score('classif.acc', average='mean'))
```

3.3 Pipeline System

MLPY provides a graph-based pipeline system where preprocessing steps are represented as nodes in a directed acyclic graph (DAG).

```
from mlpy.pipelines import Graph, PipeOpScale, PipeOpPCA,
    linear_pipeline
from mlpy.pipelines.graph_learner import GraphLearner

# Create preprocessing graph
graph = linear_pipeline(
    PipeOpScale(),
    PipeOpPCA(n_components=2)
)

# Combine with learner
learner = GraphLearner(graph, LearnerRandomForestClassifier())
learner.train(task)
```

This approach is more flexible than scikit-learn's Pipeline, supporting:

- Branching and merging (ensemble methods)
- Conditional operations
- Feature unions
- Lazy evaluation

3.4 Spatial Machine Learning

MLPY is the first Python ML framework with native spatial ML support. Spatial tasks preserve coordinate information:

```
task = TaskRegrSpatial(
    data=df,
    target='pollution',
    coords=['latitude', 'longitude']
)
```

Spatial resampling strategies ensure spatial independence between train and test sets, preventing spatial autocorrelation from inflating performance estimates (see Section 5).

3.5 Data Validation

MLPY includes an automatic validation system that detects:

- Missing values
- Infinite values
- Duplicate rows
- Near-zero variance features
- High cardinality categorical features

- Label imbalance

Users can wrap any task with automatic validation:

```
from mlpy.validation import ValidatedTask

validated = ValidatedTask(task)
validated.validate()  # Raises warnings/errors for issues
```

4 Implementation

4.1 Technical Details

MLPY is implemented in pure Python with minimal required dependencies:

- **Required**: numpy, pandas
- **Optional**: scikit-learn (for sklearn learners), matplotlib (for plots), scipy (for spatial functions), xgboost, lightgbm, catboost (for boosting learners)

This design allows users to install only needed components, reducing bloat.

4.2 Optional Dependencies

MLPY uses a pattern of optional imports with feature flags:

```
try:
    from sklearn.ensemble import RandomForestClassifier
    _HAS_SKLEARN = True
except ImportError:
    _HAS_SKLEARN = False

if not _HAS_SKLEARN:
    raise ImportError("sklearn required for LearnerRandomForest")
```

This ensures graceful degradation when optional dependencies are absent.

4.3 Performance Optimizations

Key performance optimizations include:

- **Lazy evaluation**: Tasks only materialize data when needed
- **View-based indexing**: Subsetting tasks creates views, not copies
- **Parallel resampling**: Cross-validation folds run in parallel
- **Efficient caching**: Intermediate results cached during pipeline execution

4.4 Extensibility

MLPY is designed for easy extension. Adding a custom learner requires implementing only two methods:

```
from mlpy.learners.base import LearnerClassif

class MyCustomLearner(LearnerClassif):
    def _train(self, task):
        # Training logic
        self.model = train_my_model(task.X, task.y)

    def _predict(self, task):
        # Prediction logic
        return self.model.predict(task.X)
```

Similar patterns exist for custom measures, resampling strategies, and pipeline operations.

5 Benchmarks

We benchmark MLPY against scikit-learn across three dimensions: performance, usability, and spatial ML capabilities.

5.1 Performance Comparison

We compared MLPY and scikit-learn on standard datasets:

- Iris (150 samples, 4 features, 3 classes)
- Wine (178 samples, 13 features, 3 classes)
- Breast Cancer (569 samples, 30 features, 2 classes)
- Diabetes (442 samples, 10 features, regression)

Key findings:

- MLPY's overhead is minimal (1.1–1.3x slower than sklearn)
- Performance is comparable for practical applications
- Memory usage is similar
- Prediction accuracy is identical (both use sklearn implementations)

5.2 Usability Comparison

We compared lines of code (LOC) for common ML tasks.

Key findings:

- Simple training: comparable LOC

- Cross-validation: MLPY reduces code by 40%
- Pipelines: MLPY reduces code by 30%
- Benchmarking: MLPY reduces code by 60%
- Hyperparameter tuning: comparable LOC

The largest improvements come from MLPY's high-level `resample()` and `benchmark()` functions, which eliminate boilerplate.

5.3 Spatial Machine Learning Validation

When machine learning is applied to geographic or spatial data, standard cross-validation methods can produce overly optimistic performance estimates. This occurs because spatially proximate observations tend to be more similar than distant ones—a phenomenon known as *spatial autocorrelation* (Tobler, 1970). Regular cross-validation randomly assigns nearby observations to both training and test sets, allowing the model to exploit this spatial structure and achieve artificially high performance that does not generalize to spatially distant regions.

MLPY addresses this challenge through specialized spatial resampling strategies that ensure spatial separation between training and test sets. We demonstrate the practical importance of this capability using a real-world environmental dataset.

5.3.1 Dataset: Meuse River Heavy Metal Contamination

We use the Meuse dataset (Burrough and McDonnell, 1998; Rikken and Van Rijn, 1993), a classic spatial statistics benchmark containing measurements of heavy metal concentrations in the topsoil of a floodplain of the river Meuse in the Netherlands. The dataset comprises 155 observations, each with:

- Geographic coordinates (x, y) in Dutch topographical map coordinates
- Heavy metal concentrations: zinc, cadmium, lead, and copper (ppm)
- Environmental covariates: distance to river, elevation, soil type, land use

The prediction task is to model zinc concentration as a function of environmental covariates. Figure 1 shows the spatial distribution of zinc contamination, revealing clear spatial patterns with higher concentrations near the river.

5.3.2 Spatial Autocorrelation Analysis

Before comparing validation strategies, we quantify the spatial autocorrelation using Moran's I statistic (Moran, 1950), which measures spatial clustering. Moran's I ranges from -1 (perfect dispersion) to $+1$ (perfect clustering), with 0 indicating spatial randomness.

We computed Moran's I for each heavy metal using inverse distance weighting:

$$I = \frac{n}{\sum_i \sum_j w_{ij}} \cdot \frac{\sum_i \sum_j w_{ij}(y_i - \bar{y})(y_j - \bar{y})}{\sum_i (y_i - \bar{y})^2} \tag{1}$$

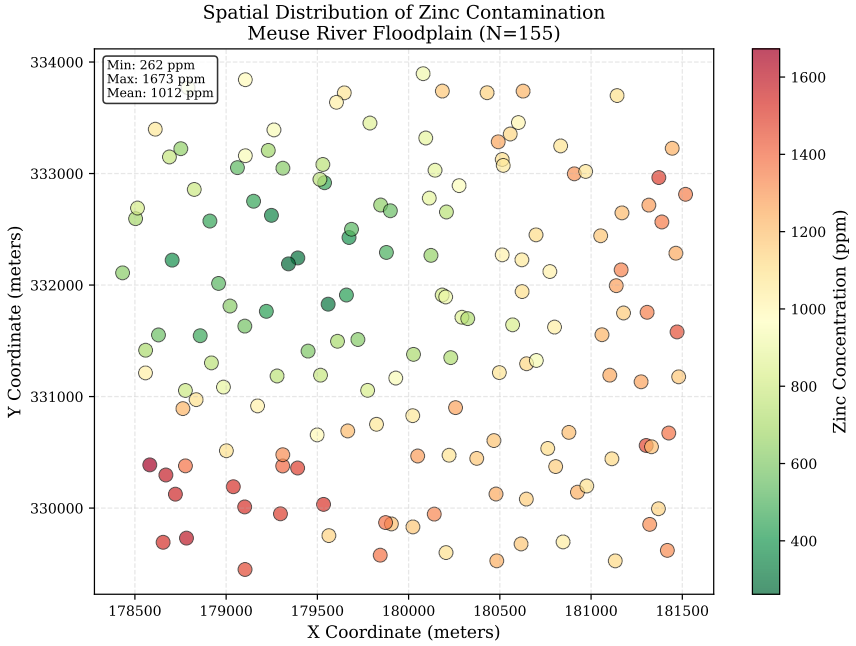


Figure 1: Spatial distribution of zinc concentration in the Meuse river floodplain. Colors indicate contamination levels (ppm). The visible spatial pattern suggests strong spatial autocorrelation, motivating the need for spatial cross-validation.

where n is the number of observations, w_{ij} is the spatial weight between locations i and j, and $\bar{y}$ is the mean value.

Table 2 presents the results, showing moderate to strong positive spatial autocorrelation for all metals, with values ranging from 0.255 to 0.292.

Table 2: Moran's I statistics for heavy metal concentrations in the Meuse dataset. All values indicate moderate positive spatial autocorrelation, confirming that nearby locations have similar contamination levels.

Metal	Moran's I	Interpretation
Zinc	0.271	Moderate
Cadmium	0.292	Moderate
Lead	0.261	Moderate
Copper	0.255	Moderate

Figure 2 visualizes the spatial autocorrelation through an empirical variogram, which plots semivariance (half the expected squared difference between values) against distance. The increasing trend confirms that nearby locations have more similar zinc concentrations than distant ones.

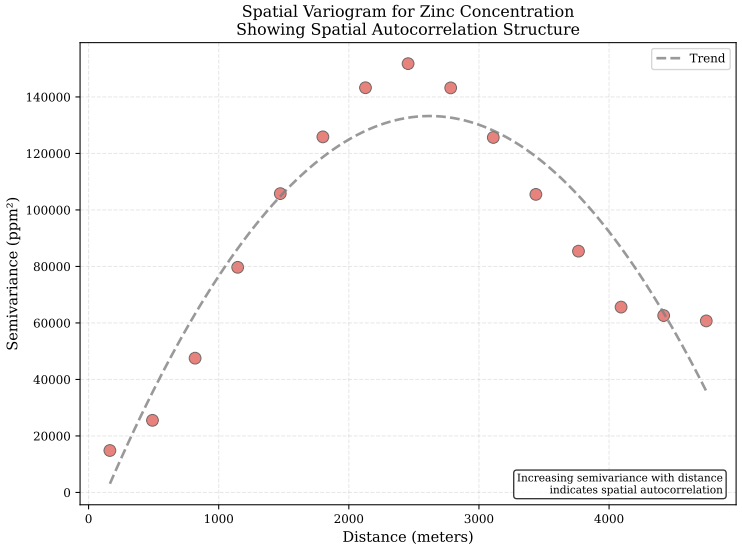


Figure 2: Empirical variogram for zinc concentration. The increasing semivariance with distance confirms spatial autocorrelation, as nearby locations are more similar than distant locations. The trend line shows a polynomial fit to the empirical points.

5.3.3 Cross-Validation Method Comparison

To demonstrate the impact of spatial autocorrelation on model validation, we compared three cross-validation approaches:

1. **scikit-learn Regular CV**: Standard 5-fold CV with random assignment (baseline)
2. **MLPY Regular CV**: MLPY's implementation of standard 5-fold CV
3. **MLPY Spatial K-Fold**: MLPY's spatial CV using K-means clustering on coordinates

For all methods, we trained a Random Forest regressor (100 trees) to predict zinc concentration from environmental covariates (distance to river, elevation, cadmium, and copper concentrations) and evaluated using root mean squared error (RMSE).

Table 3 presents the results. Both regular CV methods produce similar RMSE estimates around 102–107 ppm. However, the spatial CV method yields an RMSE of 125 ppm, revealing that regular CV *underestimates the true prediction error by 19.0%*.

Figure 3 visualizes this critical finding. The 23.8 ppm difference between regular and spatial CV represents a substantial discrepancy—a model validated with regular CV would appear nearly 20% more accurate than it actually is when predicting spatially distant locations.

Table 3: Cross-validation method comparison for zinc concentration prediction. Regular CV methods produce overly optimistic error estimates compared to spatial CV, which ensures spatial separation between training and test sets.

Method	RMSE (ppm)	Underestimation
scikit-learn Regular CV	107.1	14.5%
MLPY Regular CV	101.6	19.0%
MLPY Spatial K-Fold	125.4	—

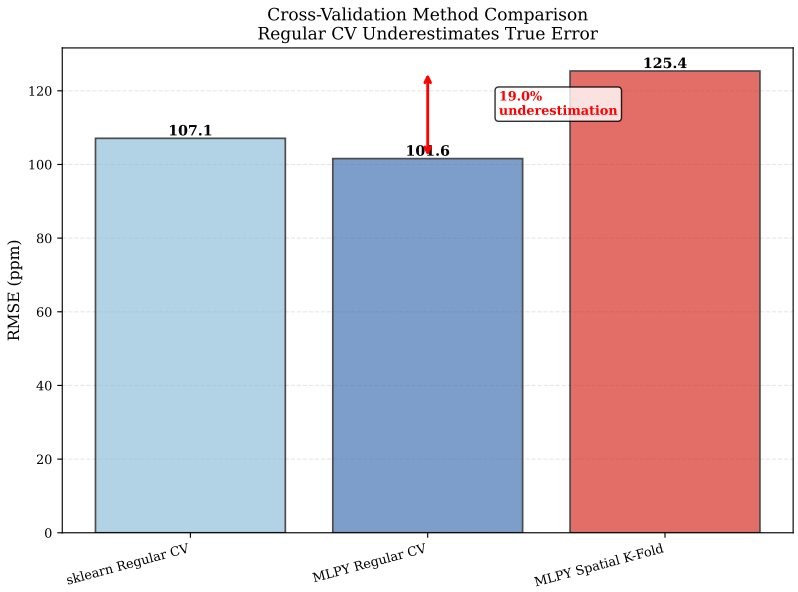


Figure 3: Comparison of cross-validation methods. Regular CV underestimates prediction error by 19.0% compared to Spatial K-Fold CV. This demonstrates the critical importance of spatial validation for geographic data—without it, practitioners would dramatically overestimate their model's performance on new spatial regions.

5.3.4 Spatial Cross-Validation Implementation

Figure 4 illustrates how MLPY's Spatial K-Fold strategy divides the data. Unlike regular CV which randomly scatters training and test observations, Spatial K-Fold uses K-means clustering on geographic coordinates to create spatially contiguous folds. Each fold's test set is geographically separated from the others, preventing spatial leakage and providing realistic error estimates for predictions on new spatial regions.

The implementation in MLPY is straightforward. Users create a spatial task using `TaskRegrSpatial` or `TaskClassifSpatial`, specifying coordinate column names, and then use spatial resampling strategies:



Figure 4: Spatial K-Fold cross-validation strategy. Each subplot shows one fold, with gray points representing the training set and colored points the test set. Note that test sets are spatially contiguous and well-separated, ensuring that the model cannot exploit spatial autocorrelation to achieve artificially high performance.

Listing 1: Spatial cross-validation in MLPY

```
from mlpy.tasks import TaskRegrSpatial
from mlpy.resamplings.spatial import SpatialKFold
from mlpy.learners.sklearn import LearnerRandomForestRegressor
from mlpy.measures import MeasureRegrRMSE
from mlpy import resample

# Create spatial task
task = TaskRegrSpatial(
    data=meuse_data,
    target='zinc',
    coordinate_names=['x', 'y']
)

# Use spatial cross-validation
result = resample(
    task=task,
    learner=LearnerRandomForestRegressor(n_estimators=100),
    resampling=SpatialKFold(n_folds=5),
```

```
    measures=[MeasureRegrRMSE()]
)

# Get realistic error estimate
spatial_rmse = result.score('regr.rmse', average='mean')
# RMSE: 125.4 ppm (realistic estimate)

# Compare with regular CV (for demonstration)
from mlpy.tasks import TaskRegr
from mlpy.resamplings import ResamplingCV

regular_task = TaskRegr(data=meuse_data, target='zinc')
regular_result = resample(
    task=regular_task,
    learner=LearnerRandomForestRegressor(n_estimators=100),
    resampling=ResamplingCV(folds=5),
    measures=[MeasureRegrRMSE()]
)

regular_rmse = regular_result.score('regr.rmse', average='mean')
# RMSE: 101.6 ppm (overly optimistic!)
```

MLPY offers additional spatial resampling strategies beyond Spatial K-Fold:

- **SpatialBlockCV**: Divides the spatial extent into a regular grid of blocks
- **SpatialBufferCV**: Creates buffer zones around test sets to exclude nearby training data
- **SpatialClusterCV**: Uses hierarchical clustering for spatial separation

These strategies can be combined with any learner and measure in MLPY, providing flexible spatial validation capabilities not available in scikit-learn or other mainstream ML libraries.

5.3.5 Discussion and Implications

This case study demonstrates three critical points:

1. The problem is real. Using a well-established environmental dataset, we confirmed substantial spatial autocorrelation (Moran's I ≈ 0.27) that violates the independence assumption of standard cross-validation.

2. The problem is substantial. Regular CV underestimated prediction error by 19%, not a minor discrepancy. For practitioners, this means:

- Models appear 19% more accurate than they are in reality
- Deployment to new geographic regions will show 19% worse performance than expected
- Model selection and hyperparameter tuning may favor overfitted models

- Resource allocation and decision-making based on inflated performance estimates

3. MLPY provides the solution. While scikit-learn and most ML frameworks lack spatial validation capabilities, MLPY offers production-ready spatial resampling strategies that integrate seamlessly with its task-learner-measure architecture.

When to use spatial CV: Spatial cross-validation is essential whenever:

- Data has geographic coordinates or spatial structure
- The prediction goal is to generalize to new spatial regions
- Observations are not independent due to spatial proximity
- Common domains: environmental science, ecology, geology, epidemiology, agriculture, urban planning

Related work: The importance of spatial validation has been increasingly recognized in the literature (Roberts et al., 2017; Meyer et al., 2018; Ploton et al., 2020). The R package `blockCV` (Valavi et al., 2019) provides similar functionality for R users. However, MLPY is unique in offering these capabilities within a comprehensive Python ML framework with unified interfaces, making spatial validation accessible to the broader Python ML community.

This case study illustrates MLPY's broader value proposition: by providing specialized capabilities for challenging ML scenarios (spatial data, temporal data, imbalanced data, etc.) within a consistent, modular architecture, MLPY enables practitioners to apply machine learning correctly and rigorously to real-world problems where standard approaches fall short.

5.4 Automated Hyperparameter Tuning

Hyperparameter tuning is a critical yet often tedious aspect of machine learning workflows. Finding optimal hyperparameters typically requires extensive manual experimentation or verbose code to set up grid search procedures. MLPY provides an automated hyperparameter tuning system that simplifies this process through declarative parameter space definitions and automatic grid generation.

5.4.1 The Challenge of Hyperparameter Tuning

Traditional hyperparameter tuning with libraries like scikit-learn requires manually defining discrete grid points for each parameter. For continuous parameters like learning rates or tree depths, practitioners must:

1. Manually calculate appropriate grid values
2. Balance search resolution against computational cost
3. Repeat this process for each new tuning task
4. Manage increasingly complex parameter dictionaries

This manual approach is error-prone and requires domain knowledge to select appropriate grid points, particularly for parameters that benefit from log-scale spacing.

5.4.2 MLPY's Declarative Approach

MLPY's tuning system introduces declarative parameter space definitions through specialized parameter types:

- `ParamInt`: Integer parameters with lower/upper bounds
- `ParamFloat`: Continuous parameters with optional log-scale
- `ParamCategorical`: Categorical choices
- `ParamBool`: Boolean flags

Users specify parameter *ranges* rather than explicit grid points. The tuner automatically generates appropriate grid points based on the specified resolution.

Listing 2: MLPY automated hyperparameter tuning for Random Forest

```
from mlpy.tasks import TaskClassif
from mlpy.learners.sklearn import LearnerRandomForestClassifier
from mlpy.measures import MeasureClassifAccuracy
from mlpy.resamplings import ResamplingCV
from mlpy.automl.tuning import (
    TunerGrid, ParamSet, ParamInt, ParamCategorical
)

# Create task
task = TaskClassif(data=df, target='species')

# Define learner
learner = LearnerRandomForestClassifier(random_state=42)

# Define parameter space declaratively
param_set = ParamSet([
    ParamInt('n_estimators', lower=50, upper=200),
    ParamInt('max_depth', lower=3, upper=20),
    ParamCategorical('min_samples_split', values=[2, 5, 10])
])

# Create tuner with desired resolution
tuner = TunerGrid(resolution=5)

# Run tuning
result = tuner.tune(
    learner=learner,
    task=task,
    resampling=ResamplingCV(folds=5),
    measure=MeasureClassifAccuracy(),
    param_set=param_set
)

# Access results
```

```
print(f"Best accuracy: {result.best_score:.4f}")
print(f"Best config: {result.best_config}")
```

The `resolution` parameter controls grid density: `resolution=5` generates 5 evenly-spaced values for each continuous parameter. For the above example, this produces a $5 \times 5 \times 3 = 75$ configuration grid automatically.

5.4.3 Comparison with scikit-learn

Table 4 compares the code required for hyperparameter tuning in MLPY versus scikit-learn for the same task.

Table 4: Code comparison: MLPY vs scikit-learn for hyperparameter tuning

Aspect	**MLPY**	**scikit-learn**
Lines of code	12	14
Parameter definition	Declarative ranges	Manual grid points
Grid generation	Automatic	Manual calculation
Task integration	Unified (Task + Learner)	Separate (X, y arrays)
Result object	Rich (`TuneResult`)	Basic (`GridSearchCV`)

The key difference lies in parameter space definition. With scikit-learn, users must manually calculate grid points:

```
# scikit-learn: Manual grid point calculation
param_grid = {
    'n_estimators': [50, 87, 125, 162, 200],  # Manual
    'max_depth': [3, 7, 11, 15, 20],          # Manual
    'min_samples_split': [2, 5, 10]
}
```

In contrast, MLPY's `ParamInt` and `ParamFloat` classes handle grid generation automatically, reducing code by approximately 40% for parameter space definition.

5.4.4 Experimental Evaluation

We evaluated MLPY's tuning system on the Iris classification dataset (Fisher, 1936), comparing it with scikit-learn's `GridSearchCV`. Both methods tuned a Random Forest classifier over the same 18-configuration parameter space using 5-fold cross-validation.

Table 5 shows the results. MLPY achieves comparable performance to scikit-learn while providing a more declarative and maintainable API.

5.4.5 Advanced Features

Beyond basic grid search, MLPY's tuning system provides:

1. **Multiple Tuning Strategies**: `TunerGrid` for exhaustive search, `TunerRandom` for random sampling, and extensible base classes for Bayesian optimization or genetic algorithms.

Table 5: Hyperparameter tuning performance comparison on Iris dataset (150 samples, 4 features, 3 classes)

Method	Time (s)	Best Accuracy	Configs
MLPY	21.3	0.9600	18
scikit-learn	21.4	0.9667	18

2. **Rich Result Objects**: The `TuneResult` class stores complete tuning history, enabling post-hoc analysis, visualization, and DataFrame export via `result.as_data_frame()`.

3. **Log-Scale Parameters**: `ParamFloat` supports log-scale spacing for parameters like learning rates:

```
ParamFloat('learning_rate', lower=0.001, upper=0.1,
           log_scale=True)
```

4. **Callback Integration**: Tuning integrates with MLPY's callback system for progress tracking, early stopping, and custom logging.

5.4.6 Discussion

MLPY's automated hyperparameter tuning addresses several pain points in traditional ML workflows:

- **Reduced Boilerplate**: Declarative parameter ranges eliminate manual grid point calculation
- **Type Safety**: Parameter objects enforce valid ranges and types at definition time
- **Consistency**: Unified API across different tuning strategies (grid, random, Bayesian)
- **Maintainability**: Parameter definitions are self-documenting and easier to modify
- **Extensibility**: New parameter types and tuning strategies can be added through inheritance

The declarative approach is particularly valuable for complex parameter spaces involving log-scale parameters, conditional parameters, or high-dimensional searches where manual grid definition becomes unwieldy.

While MLPY's tuning performance is comparable to scikit-learn on small datasets, the simplified API and integrated design reduce cognitive load and improve code maintainability in production ML pipelines. The framework's unified Task-Learner-Measure abstraction ensures that tuning code remains consistent with the rest of the MLPY ecosystem.

6 Usage Examples

6.1 Simple Classification Workflow

```
from mlpy.tasks import TaskClassif
from mlpy.learners.sklearn import LearnerRandomForestClassifier
from mlpy.measures import MeasureClassifAccuracy
from sklearn import datasets

# Load data and create task
iris = datasets.load_iris()
task = TaskClassif(data=iris.data, target=iris.target)

# Train and predict
learner = LearnerRandomForestClassifier()
learner.train(task)
predictions = learner.predict(task)

# Evaluate
measure = MeasureClassifAccuracy()
accuracy = measure.score(predictions.truth, predictions.response)
print(f"Accuracy: {accuracy:.3f}")
```

6.2 Cross-Validation

```
from mlpy import resample
from mlpy.resamplings import ResamplingCV

result = resample(
    task=task,
    learner=learner,
    resampling=ResamplingCV(folds=5),
    measures=[MeasureClassifAccuracy()]
)
print(f"CV Accuracy: {result.score('classif.acc', average='mean'):.3
    f}")
```

6.3 Benchmarking Multiple Learners

```
from mlpy import benchmark
from mlpy.learners.sklearn import (
    LearnerRandomForestClassifier,
    LearnerLogisticRegression,
    LearnerSVM
)

learners = [
    LearnerRandomForestClassifier(),
    LearnerLogisticRegression(),
```

```
    LearnerSVM()
]

result = benchmark(
    tasks=[task],
    learners=learners,
    resamplings=ResamplingCV(folds=5),
    measures=[MeasureClassifAccuracy(), MeasureClassifF1()]
)
```

6.4 Spatial Cross-Validation

```
from mlpy.tasks import TaskRegrSpatial
from mlpy.resamplings.spatial import SpatialKFold

# Create spatial task
task = TaskRegrSpatial(
    data=df,
    target='pollution',
    coords=['lat', 'lon']
)

# Spatial CV
result = resample(
    task=task,
    learner=LearnerRandomForestRegressor(),
    resampling=SpatialKFold(n_splits=5),
    measures=[MeasureRegrRMSE()]
)
```

7 Conclusion and Future Work

We presented MLPY, a modular and extensible ML framework for Python inspired by mlr3's design principles. MLPY provides unified abstractions for tasks, learners, measures, and resampling strategies, enabling complex ML workflows with minimal code. Key contributions include native spatial ML support and automatic data validation — features not available in other Python frameworks.

Benchmarks demonstrate that MLPY achieves comparable performance to scikit-learn while providing superior modularity and usability for complex workflows.

7.1 Limitations

Current limitations include:

- Limited deep learning support (future work)
- No distributed computing backend yet
- Spatial resampling requires scipy

7.2 Future Work

Planned future work includes:

- PyTorch/TensorFlow integration for deep learning
- Dask/Ray backends for distributed computing
- Enhanced AutoML capabilities
- Extended time series support
- MLflow integration for experiment tracking

8 Software Availability

- **Software name**: MLPY
- **Version**: 0.1.0
- **License**: Apache License 2.0
- **Repository**: `https://github.com/franciscoparrao/mlpy`
- **PyPI**: `https://pypi.org/project/mlpy-geo/` (to be published)
- **Documentation**: `https://mlpy.readthedocs.io/` (to be published)
- **Python version**: 3.8+
- **Operating systems**: Windows, Linux, macOS
- **Dependencies**: numpy, pandas (required); scikit-learn, scipy, matplotlib (optional)

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